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Runoff sensitivity to snow depletion curve representation within a continental scale hydrologic model

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Abstract

The spatial variability of snow water equivalent (SWE) can exert a strong influence on the timing and magnitude of snowmelt delivery to a watershed. Therefore, the representation of sub-grid or sub-watershed snow variability in hydrologic models is important for accurately simulating snowmelt dynamics and runoff response. The U.S. Geological Survey National Hydrologic Model infrastructure with the precipitation-runoff modelling system (NHM-PRMS) represents the sub-grid variability of SWE with snow depletion curves (SDCs), which relate snow-covered area to watershed-mean SWE during the snowmelt period. The main objective of this research was to evaluate the sensitivity of simulated runoff to SDC representation within the NHM-PRMS across the continental United States (CONUS). SDCs for the model experiment were derived assuming a range of SWE coefficient of variation values and a lognormal probability distribution function. The NHM-PRMS was simulated at a daily time step for each SDC over a 14-year period. Results highlight that increasing the sub-grid snow variability (by changing the SDC) resulted in a consistently slower snowmelt rate and longer snowmelt duration when averaged across the hydrologic response unit scale. Simulated runoff was also found to be sensitive to SDC representation, as decreases in simulated snowmelt rate by 1 mm day^{-1} resulted in decreases in runoff ratio by 1.8% on average in snow-dominated regions of the CONUS. Simulated decreases in runoff associated with slower snowmelt rates were approximately inversely proportional to increases in simulated evapotranspiration. High snow persistence and peak SWE:annual precipitation combined with a water-limited dryness index was associated with the greatest runoff sensitivity to changing snowmelt. Results from this study highlight the importance of carefully parameterizing SDCs for hydrologic modelling. Furthermore, improving model representation of snowmelt input variability and its relation to runoff generation processes is shown to be an important consideration for future modelling applications.

KEYWORDS
continental United States, model sensitivity, precipitation-runoff modelling system, scaling, snow depletion curve, snowmelt, streamflow, sub-grid variability

1 | INTRODUCTION

Seasonal snow is a critical component of the surface energy balance and hydrologic cycle in mountainous areas and high latitudes; changes to snowmelt duration, timing and magnitude in these regions may substantially alter hydrological processes. Mountains, which function as essential water towers, generally produce more streamflow per unit area than lower lying terrain (Christensen, Wood, Voisin, Lettenmaier, & Palmer, 2004; Hunsaker, Whitaker, & Bales, 2012; Viviroli, Durr, Messerli, Meybeck, & Weingartner, 2007), so hydrologic changes in these areas can have substantial effects both where snowmelt occurs as well as across large areas they feed downstream. Approximately 2 billion people are expected to experience diminished water supplies due to seasonal snowpack decline this century (Barnett, Adam, & Lettenmaier, 2005; Mankin, Viviroli, Singh, Hoekstra, & Diffenbaugh, 2015). Snowpacks have declined across the western United States (U.S.) in recent years (Clow, 2010; Fritze, Stewart, & Pebesma, 2011; Harpold et al., 2012; Mote, Hamlet, Clark, & Lettenmaier, 2005; Mote, Li, Lettenmaier, Xiao, & Engel, 2018; Regonda, Rajagopalan, Clark, & Pitlick, 2005; Stewart, Cayan, & Dettinger, 2005), with both empirical and modelled evidence that snowpack decline may lead to reduced runoff efficiency (ratio of total runoff to total precipitation; Barnhart et al., 2016; Berghuijs, Woods, & Hrachowitz, 2014; Clow, 2010; Furey, Kampf, Lanini, & Dozier, 2012; Hammond, Harpold, Weiss, & Kampf, 2019; Jefferson, 2011; Regonda et al., 2005; Stewart, Cayan, & Dettinger, 2004; Stewart et al., 2005; Xiao, Udall, & Lettenmaier, 2018). However, hydrologic response to snowpack decline is heterogeneous; in some areas, snowpack reductions produce declines in streamflow, but this does not occur everywhere (McCabe, Wolock, & Valentin, 2018). Sources of this variability in hydrologic response are often unknown and likely a result of interactions between climatic, vegetative, topographic and edaphic factors with the relative importance of each factor changing between climatic zones and along elevation gradients within each climatic zone. Therefore, physically based hydrologic models that are run with consistent methods across broad extents and can be used to evaluate hydrologic processes and dynamics that vary considerably across spatial and temporal scales may provide important insights into the relative sensitivities of hydrologic responses to these anticipated snowpack declines.

The seasonal snowpack exhibits marked variability both spatially and temporally across the landscape (Lopez-Moreno et al., 2015), which exerts a strong influence on the timing and magnitude of snowmelt delivery to a watershed (Anderton, White, & Alvera, 2002; Liston, 1999; Luce, Tarboton, & Cooley, 1998) and its streamflow response (DeBeer & Pomeroy, 2017; Lundquist & Dettinger, 2005; Lundquist, Dettinger, & Cayan, 2005). Therefore, the representation of the sub-grid or sub-watershed snow variability in broad-scale hydrologic models is particularly important for accurately simulating variations in energy fluxes, snowmelt dynamics and runoff response (Clark et al., 2011; He, Ohara, & Miller, 2019; Liston, 1999, 2004). Snow depletion curves (SDCs) that relate the snow-covered area (SCA) to the mean snow water equivalent (SWE) for a given hydrologic response unit (HRU) are often used to represent the sub-grid variability of snowmelt processes within hydrologic models (Anderson, 1968; Driscoll, Hay, & Bock, 2017; Liston, 1999; Luce & Tarboton, 2004; Magand, Ducharne, Le Moine, & Gascoin, 2014; Markstrom et al., 2015; Martinec & Rango, 1981; Yang, Dickinson, Robock, & Vinnikov, 1997). Luce and Tarboton (2004) highlight that dimensionless SDCs can show good inter-annual agreement, but also suggest there may be environments or scales where dimensionless SDCs may change from year to year. Furthermore, SDCs are difficult to derive over large regions (Driscoll et al., 2017; Fassnacht et al., 2016; Shamir & Georgakakos, 2007), variable in their representation between models (Essery & Pomeroy, 2004; Liston, 2004), and often applied consistently over heterogeneous landscapes. Where more realistically, the representation of the SWE-SCA relation should be regionally variable (Driscoll et al., 2017), with differences between landscape type and melt energy accounted for by the HRUs (DeBeer & Pomeroy, 2009, 2010; Donald, Soulis, Kouwen, & Pietroniro, 1995; Dornes, Pomeroy, Pietroniro, Carey, & Quinton, 2010), that change based on the scale of interest (Blöschl, 1999; Seyfried & Wilcox, 1995). Given the importance of scaling and spatial representation of SDCs described here, it is important to understand the sensitivity of simulated hydrologic fluxes to SDC representation at a scale of interest.

In a review of past approaches to represent sub-grid SWE variability in modelling, Clark et al. (2011) suggest that the most physically realistic approach is using a probability distribution function (pdf), as it only relies on an estimate of the coefficient of variation (CV; ratio of the standard deviation to the mean) of SWE, an observable property in the field (e.g., Winstral & Marks, 2014). Various studies have highlighted that snow distributions can be reasonably described by a lognormal pdf (DeBeer & Pomeroy, 2009; Donald et al., 1995; Faria, Pomeroy, & Essery, 2000; Pomeroy et al., 1998), whereas others have suggested that snow distributions tend to more closely follow a gamma pdf (Egli, Jonas, Grunewald, Schirmer, & Burlando, 2012; Skaugen, 2007; Winstral & Marks, 2014). Despite these differences, Luce and Tarboton (2004) highlighted that the representation of sub-grid snow variability is much more sensitive to the distribution parameters (e.g., CV of SWE and mean SWE) as opposed to distribution type. Liston (2004) developed a global classification scheme of the CV of SWE and showed how a lognormal pdf could be used to simulate the sub-grid variability of SWE and associated snowmelt by assuming uniform melting conditions, offering an approach to represent the sub-grid snow variability across broad-scale modelling applications. Although the assumption of uniform melting conditions is likely violated in many areas and is dependent on the HRU scale, Luce et al. (1998) noted that most of the spatial variability in SWE at the watershed scale is due to variations in accumulation, rather than variations in melt. Based on these sub-grid SWE variability approaches, this study used a range of SDCs derived by a lognormal pdf to perform a sensitivity analysis of simulated hydrologic fluxes.

The U.S. Geological Survey (USGS) Water Availability and Use Science Program (https://www.usgs.gov/water-resources/water-availability-and-use-science-program) aims to quantify the availability of water resources at the watershed scale for the U.S. Towards this goal, the national-extent application of the National Hydrologic Model infrastructure with the precipitation-runoff modelling system (NHM-PRMS) (Regan et al., 2018; Regan et al., 2019) has been applied to simulate water budget components contributing to streamflow. Although empirical studies have shown location-specific results of the importance of the timing and duration of snowmelt to streamflow (e.g., Anderton et al., 2002; DeBeer & Pomeroy, 2017; Luce et al., 1998), this study investigates the sensitivity of simulated hydrologic component fluxes to the representation of snowmelt rate in the NHM-PRMS across the continental United States (CONUS). More specifically, a model experiment was completed with the goal of assessing the sensitivity of simulated runoff to the representation of watershed-scale snowmelt rate, as represented by SDCs derived by a lognormal pdf, within the model. Model simulations with the same meteorological forcing and model parameters but different SDCs allowed for the direct comparison of simulated hydrological fluxes between simulations. The objectives of this study were to (a) determine which regions of the CONUS are relatively more or less sensitive to a change in snowmelt representation in the model, and (b) evaluate how the simulated changes in hydrologic fluxes and partitioning are related to runoff sensitivity. Results from this study will help to better understand the importance and sensitivity of NHM-PRMS SDC representation across the CONUS with the goal of focusing future investigations to further improve the USGS's national modelling and predictive capacity of water availability for the U.S.

2 | METHODS

2.1 | Model description

The USGS NHM-PRMS was applied in this study (Regan et al., 2018; Regan et al., 2019). The PRMS is a daily time-step, deterministic, distributed-parameter, process-based hydrologic model used to simulate hydrologic component fluxes in response to precipitation and climate forcing (Markstrom et al., 2015). The NHM-PRMS was applied across the extent of the CONUS on the Geospatial Fabric (Viger & Bock, 2014) composed of 109,951 HRUs that represent a given watershed contributing area to a given stream segment divided into left and right banks (Regan et al., 2018; Regan et al., 2019; Viger & Bock, 2014). The mean ± standard deviation (median) HRU size of the Geospatial Fabric is 75 ± 399 km^2 (33 km^2). Regan et al. (2018) provide a detailed description of the Geospatial Fabric HRU delineation procedures, which were based on points of interest discussed with the hydrologic modelling community; optimizing HRU delineations specifically based on the representation of the spatial heterogeneity of snow dynamics was not feasible during the Geospatial Fabric development. The NHM-PRMS utilizes nationally consistent hydrologic model development (i.e., physical and statistical methods for discretization, characterization, parameterization, forcing, calibration and evaluation) for simulating watershed-scale hydrologic processes across the CONUS (Regan et al., 2018).

The NHM-PRMS was calibrated for each HRU based on five baseline calibration datasets derived from CONUS-scale remote sensing and/or modelled products with error bounds at daily, monthly, or annual time scales (Hay, 2019). These baseline calibration datasets included simulated runoff (Bock, Farmer, & Hay, 2018; Bock, Hay, McCabe, Markstrom, & Atkinson, 2016), simulated and remotely sensed actual evapotranspiration (Bock et al., 2016; Running, Mu, & Zhao, 2017; Senay et al., 2013), empirical regression estimates and simulated recharge (Doll, Mueller Schmied, Schuh, Portmann, & Eicker, 2014; Reitz, Sanford, Senay, & Cazenas, 2017), simulated soil moisture (Kalnay et al., 1996; Reichle et al., 2011; Xia et al., 2012) and remotely sensed SCA (Hall & Riggs, 2016b). A Fourier amplitude sensitivity test (FAST; Cukier, Fortuin, Shuler, Petschek, & Schaibly, 1973; Markstrom, Hay, & Clark, 2016; Saltelli et al., 2006; Schaibly & Shuler, 1973) was used to assess how the parameters chosen for calibration were related to overall sensitivity of each baseline calibration dataset and thus was used to determine the step-wise order of calibration for each HRU (Hay, 2019). The optimal set of calibration parameters for each HRU was chosen based on a multiple-objective, step-wise, automated calibration procedure (Hay & Umemoto, 2007) using the shuffled complex evolution (SCE) global search algorithm (Duan, Gupta, & Sorooshian, 1993; Duan, Sorooshian, & Gupta, 1992; Duan, Sorooshian, & Gupta, 1994). The objective function for each step in the by-HRU calibration used the normalized root mean square error. Hay (2019) provides additional detail and the baseline HRU calibration of the NHM-PRMS that was used in this study.

PRMS uses a dimensionless SDC to simulate the fraction of SCA associated with simulated SWE normalized by the threshold SWE value (SWE_{100}; the maximum SWE amount for a given HRU below which the SDC is applied as further decreases in SWE are associated with reductions in SCA) within an HRU during the snowmelt period (Anderson, 1968; Markstrom et al., 2015). Hay (2019) calibrated the NHM-PRMS SDCs at the HRU level using SCA observations from the MODIS satellite (MOD10C1; Hall & Riggs, 2016b). In this study, we used the calibrated baseline version of the NHM-PRMS that is provided by Hay (2019) for our model experiment, with the exception of the calibrated SDCs (described in the next section), to evaluate the sensitivity of simulated runoff to SDC representation within the model. An important scaling and model structure consideration to this study is how hydrologic models distribute simulated snowmelt input across an HRU. In PRMS, snowmelt input from the SCA is uniformly broadcasted across the entire HRU rather than being concentrated within the SCAs only. Therefore, there is potential scaling uncertainty in relating the watershed-wide surface water input rate to runoff generation processes that are spatially dependent on snowmelt variability. Based on this potential uncertainty, we evaluate the importance of model structure as it relates to the results of this study in the scaling and model structure considerations section later in this paper.

2.2 | Model experiment

The model experiment completed for this study consisted of seven NHM-PRMS model simulations. For each of the model simulations, we replaced the HRU-calibrated NHM-PRMS SDCs with a single SDC derived by assuming a lognormal pdf and assigning a CV value. Luce, Tarboton, and Cooley (1999) described how a dimensionless SDC could be estimated for a given pdf by evaluating the decreases in SCA with decreasing normalized watershed-mean SWE for several values of watershed-mean snowmelt amount (assuming uniform snowmelt conditions). As summarized by Donald et al. (1995), Luce et al. (1999), Essery and Pomeroy, Essery, and Toth (2004), and Clark et al. (2011), these values can be evaluated based on the pdf with the following equations:

$$\mathrm{SCA} = \int_M^\infty f(S)dS \tag{1}$$

$$\overline{\mathrm{SWE}} = \int_M^\infty (S-M)f(S)dS, \tag{2}$$

where SCA is the fractional snow-covered area, $\overline{\mathrm{SWE}}$ is the grid-mean SWE, M is the total melt since the beginning of the snow season and S is the SWE (as described by the pdf). Liston (2004) and DeBeer and Pomeroy (2009) described this framework with the use of a two-parameter lognormal pdf:

$$f(S) = \frac{1}{S\zeta\sqrt{2\pi}}\exp\left\{-\frac{1}{2}\left[\frac{\ln(S)-\lambda}{\zeta}\right]^2\right\}, \tag{3}$$

with

$$\lambda = \ln(\mu) - \frac{1}{2}\zeta^2 \tag{4}$$

$$\zeta^2 = \ln\left(1+\mathrm{CV}^2\right), \tag{5}$$

where λ and ζ are the distribution parameters related to the mean (μ) and CV of the pre-melt SWE (S). Using this technique, we derived seven dimensionless SDCs assuming a two-parameter lognormal pdf with CV values ranging from 0.1 to 2.0 (Figure 1). Because this technique assumes uniform snowmelt conditions, the derivation of the SDCs did not change when changing the mean pre-melt SWE or snowmelt rate and were only dependent on the CV value. Each SDC represents how the aerial SCA of an HRU decreases with the fractional decrease of SWE below the threshold SWE value (SWE_{100}) for the given HRU (Donald et al., 1995; Fassnacht et al., 2016; Markstrom et al., 2015). As described above, the simulated snowmelt input from the SCA in PRMS is uniformly broadcasted across the entire HRU. Additionally, in PRMS, new snowfall that occurs during the snowmelt period will increase the SCA to its maximum value and a linear interpolation is used to between this value and the previous location on the depletion curve as the new snowfall melts (Figure 1b; Markstrom et al., 2015).

The seven NHM-PRMS simulations were completed at a daily time step over a 14-year period (2003–2016) using daily minimum and maximum air temperature and daily precipitation forcing data from Daily Surface Weather and Climatological Summaries (DAYMET, https://daymet.ornl.gov/) that were spatially aggregated from the gridded DAYMET dataset (Thornton et al., 2016) using the USGS GeoData Portal (GDP, http://cida.usgs.gov/gdp/; Blodgett, Booth, Kunicki, Walker, & Viger, 2011). Each of the seven model experiment simulations used the baseline HRU calibrated NHM-PRMS model parameters from Hay (2019), replacing all SDCs with those derived from SWE CV values ranging from 0.1 to 2.0 (Figure 1). This model experiment allowed for nationally consistent methods to evaluate the sensitivity of simulated runoff to varying SDC representations.

Model output variables from each simulation that were evaluated in this study are described in Table 1. To evaluate simulation results

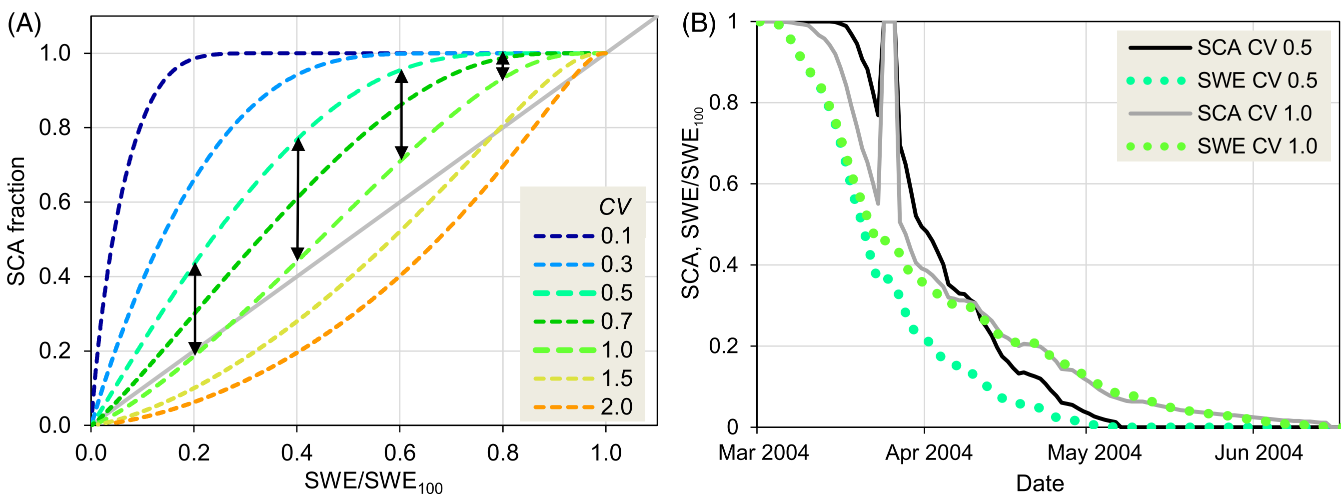

FIGURE 1 (a) Snow depletion curves (SDCs) derived based on a lognormal probability distribution function and snow water equivalent (SWE) coefficient of variation (CV) value used for the seven model experiment simulations. The vertical arrows denote the difference between the CV 0.5 and 1.0 simulations. (b) Example illustration of the depletion of snow-covered area (SCA) and SWE during the snowmelt period based on the CV 0.5 and CV 1.0 SDCs that highlight how the simulated SCA responds to new snowfall during the snowmelt period. SWE_{100} is the maximum SWE amount for a given hydrologic response unit below which the SDC is applied, as further decreases in SWE are associated with reductions in SCA

over a consistent decadal mean condition, annual summary statistics were computed from daily model outputs for every HRU, and a mean was computed of annual summary statistics over the 14-year period. To directly evaluate the sensitivity of the model simulation results to the change in SDC, a linear regression between the CV value (explanatory variable) and model output summary value (response variable) was computed for each HRU. We define the model output (e.g., runoff ratio; Table 1) sensitivity as the slope of the linear regression over a change in CV of 1.0. Model output data that support the findings of this study are available in Sexstone and Driscoll (2019).

The snow persistence (SP) metric derived from MODIS SCA (MOD10A2; Hall & Riggs, 2016a) and described in detail by Hammond et al. (2018) was used in this study to classify HRUs with similar snow properties and evaluate runoff sensitivity to the choice of SDC (Table 1). Annual SP layers were obtained for water years 2001 to 2015 (Hammond, Saavedra, & Kampf, 2017) and averaged across all years to obtain mean annual SP. Mean annual SP was then zonally averaged for each HRU with full SP grid coverage (Figure S1) and HRUs were then classified into the following four categories: low snow $SP \leq 0.25$, intermittent snow $0.25 < SP \leq 0.50$, transitional snow $0.50 < SP \leq 0.75$ and persistent snow $SP > 0.75$. Non-parametric Kruskal-Wallis and Wilcoxon rank sum analyses

TABLE 1 Definitions of model variables and metrics used in this study

Variable	Definition
Snow water equivalent (SWE)	Snowpack water equivalent on each hydrologic response unit (HRU)
Snow-covered area (SCA)	SCA on each HRU
Snowmelt	Snowmelt from snowpack on each HRU
Runoff ratio	Total flow leaving each HRU normalized by precipitation
Evaporative index	Actual evapotranspiration (ET) for each HRU normalized by precipitation
SWE precipitation ratio (SWE:P)	Maximum SWE on each HRU normalized by precipitation
Dryness index (PET/P)	Potential ET for each HRU normalized by precipitation
Surface runoff efficiency	Surface runoff to the stream network for each HRU normalized by precipitation
Interflow efficiency	Interflow from gravity and preferential-flow reservoirs to the stream network for each HRU normalized by precipitation
Groundwater efficiency	Groundwater discharge from groundwater reservoirs to the stream network for each HRU normalized by precipitation
Snow persistence (SP)	Fraction of time snow is present on the ground between January and July for each HRU based on MODIS SCA satellite observations.

Note: Refer to Markstrom et al. (2015) for further description of precipitation-runoff modelling system variables and Hammond, Saavedra, and Kampf (2018) for further description of the snow persistence metric.

(McDonald, 2014) were then used to test for significant differences (p-values <.01) in aridity and multiple metrics of runoff efficiency between snow zones.

3 | RESULTS

3.1 | Differences in variable response to SDC representation

3.1.1 | Simulated snowmelt

Simulated peak SWE was relatively unchanged between the model experiment simulations (Figure 2a), as the varying SDCs only resulted in changes to the simulated snowmelt period. Deviations in simulated peak SWE were only observed in HRUs where the maximum SWE occurred following the initiation of snowmelt in a given year. More substantial differences between model simulations were observed in simulated snowmelt rate (Figure 2b) and simulated snowmelt duration (Figure 2c). With an increase in the SWE CV and its associated SDC (Figure 1), simulated snowmelt rate decreased and simulated snowmelt duration increased monotonically (Figure 2). In other words, increasing the sub-grid snow variability as represented by the SDC resulted in a consistently slower simulated snowmelt rate and longer simulated snowmelt season at the HRU scale across the CONUS. Total simulated snowmelt amount also increased with increasing snow variability (Figure 2d), as the longer simulated snowmelt duration yielded a greater period for precipitation as rain or snow to fall onto the existing snowpack and be counted towards simulated snowmelt amount. The sensitivity of snowmelt rate to the SDC representation (i.e., the regression slope between the CV value and snowmelt rate for each HRU) highlights that changes in snowmelt rate were widespread across regions of the CONUS that receive snow and were greatest in snow-dominated areas such as the western U.S. mountains (Figure 3).

3.1.2 | Simulated runoff

The sensitivity in runoff ratio across all model simulations highlights widespread decreases in overall runoff with increases in CV because of slower snowmelt conditions at the HRU scale (Figure 4). Although the mean decrease in runoff ratio across all CONUS HRUs was equal to −0.5%, runoff ratio decreased by as much as 12% in highly sensitive regions (Figure 4). Runoff ratio sensitivity to changing snowmelt rate at the HRU scale was greatest in the snow-dominated western U.S. mountains (Figure 4). Additionally, little to no simulated runoff sensitivity in much of the central and eastern U.S. (except for the most northern regions) was observed (Figure 4) despite these regions receiving winter snow accumulation. A comparison between simulated evaporative index (including all evaporation, transpiration and sublimation processes) and simulated runoff ratio shows that changes in runoff ratio were approximately inversely proportional to changes in evaporative index during the model experiment (Figure 4). A linear

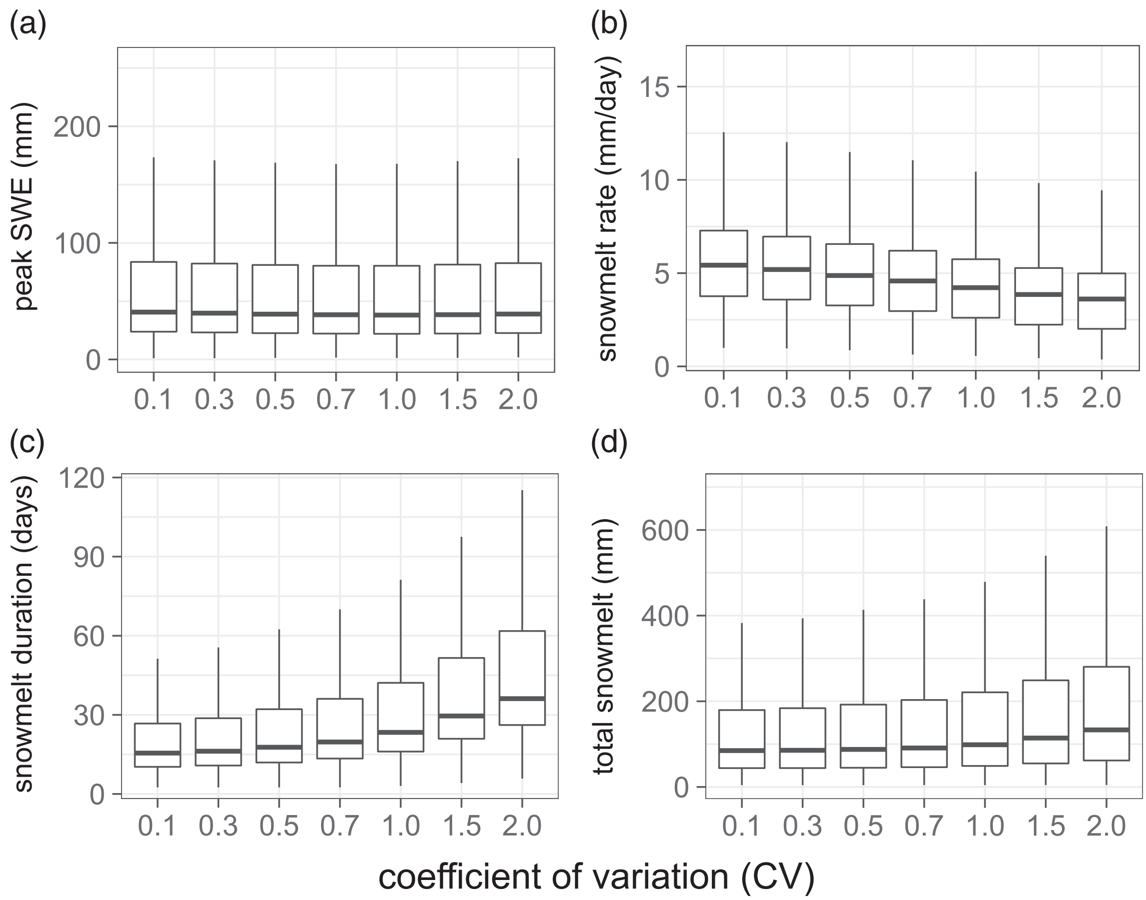

FIGURE 2 Boxplots of (a) peak snow water equivalent, (b) snowmelt rate, (c) snowmelt duration and (d) total snowmelt for each of the coefficient of variation model experiment simulations showing the median (black horizontal line), 25th and 75th percentiles (box), and the largest (smallest) value within 1.5 times the interquartile range above (below) the 75th (25th) percentile (whiskers)

regression between the simulated evaporative index sensitivity and the simulated runoff ratio sensitivity for runoff ratio decreases greater than 5% (n = 2,458; runoff ratio sensitivity = −0.923 × evaporative index sensitivity – 0.004; R^2 = 0.96; p-value <.001) indicates that approximately 92% of the decreases in runoff ratio may be attributed to increases in evaporative index. Based on the hydrologic processes simulated within the NHM-PRMS (Markstrom et al., 2015), this result indicates that increases in simulated groundwater flow to deep groundwater storage may account for only approximately 8% of the simulated runoff ratio decreases. Hydrologic partitioning between surface runoff, interflow and groundwater (Table 1) also displayed widespread sensitivity to changing snowmelt rate and duration (Figures S2 and S3); however, these changes did not always translate into overall runoff sensitivity (Figure 4c).

3.2 | Regional sensitivity

3.2.1 | Western U.S. snow amount and aridity

Given that runoff ratio sensitivity was greatest in snow-dominated regions of the western U.S. (Figure 4c), we evaluated the relation of runoff sensitivity to varying snow metrics within this region. First, we compared runoff sensitivity to mean SP (%) computed from MODIS SCA observed during the simulation period (Figure S1;Hammond et al., 2017; Hammond et al., 2018). A statistically significant difference (p-value <.001) in simulated runoff ratio sensitivity was observed between the low snow, intermittent, transitional and persistent snow zones using the non-parametric Kruskal-Wallis test (Figure 5). The runoff ratio sensitivity across all model simulations was nonlinearly related to SP, with greater decreases in runoff ratio associated with higher SP (Figure 5). The spread in the distribution of runoff ratio sensitivities was greater for transitional and persistent SP zones as compared to the low snow and intermittent zones (Figure 5). The simulated runoff ratio sensitivity was also compared with snow metrics simulated by the NHM-PRMS. The sensitivity in simulated snowmelt rate (Figure 3b) was significantly linearly related (R^2 = 0.60; p-value <.001) with the simulated runoff ratio sensitivity (Figure 6a). This relation highlights that increases (decreases) in simulated snowmelt rate by 1 mm day^{-1} resulted in increases (decreases) in runoff ratio 1.8% on average in snow-dominated regions of the CONUS. Additionally, the ratio of simulated peak SWE to annual precipitation (SWE:P; Figure S4) for the western U.S. was significantly linearly related (R^2 = 0.83; p-value <.001) with the runoff ratio sensitivity (Figure 6b).

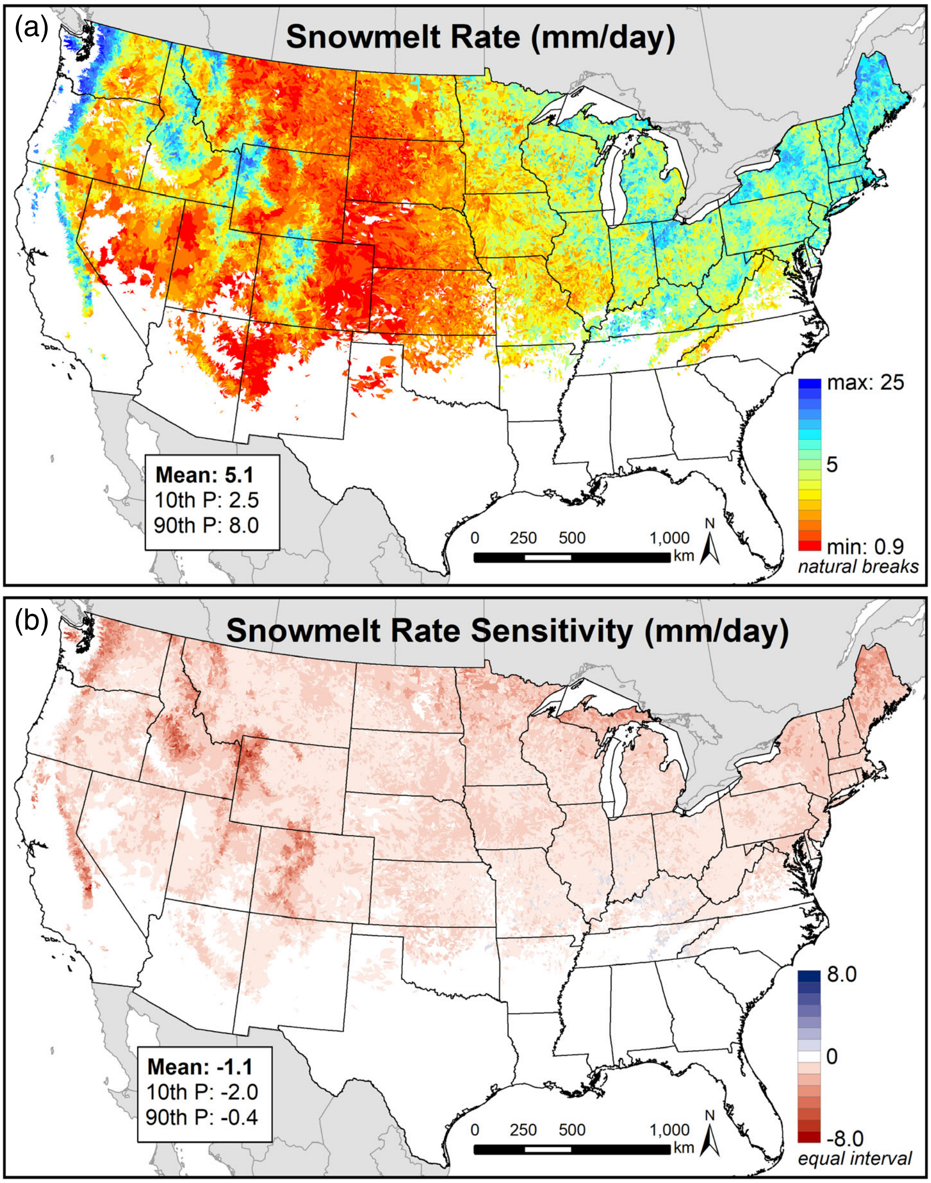

FIGURE 3 Spatial variability of the (a) snowmelt rate simulated for the coefficient of variation 0.5 model simulation and (b) the snowmelt rate sensitivity computed across all model simulations. Mean value and 10th and 90th percentiles (P) across all hydrologic response units are shown in each panel

The sensitivity in simulated runoff ratio was additionally found to be related to aridity in the western U.S. region. Dryness index (PET/P; Table 1) is a measure of the relative water limitation (dryness index >1) or energy limitation (dryness index <1) of an HRU (Budyko, 1974; Knowles et al., 2015). The simulated dryness index was divided into six quantiles for each of the western U.S. SP zones (e.g., Figure 5) to evaluate its effect on overall runoff sensitivity (Figure 7). Results highlight a statistically significant difference (p-value <.001; Kruskal-Wallis test) between the runoff ratio sensitivities divided by dryness index quantiles for each SP zone (Figure 7). A similar pattern for intermittent, transitional and persistent snow zones showed the greatest runoff sensitivity occurring for dryness index values with water limitation between approximately 1 and 2. As dryness index increased above this range, the overall runoff sensitivity decreased. Likewise, runoff sensitivity also decreased for energy limited HRUs with dryness index values <1. This pattern highlights the importance of aridity to the runoff sensitivity; however, this pattern also highlights that HRUs with the highest water limitation likely receive less snowfall and are thus less sensitive to changing snowmelt than regions that receive more snow.

3.2.2 | Regional sensitivity of runoff in snow-dominated western U.S.

We evaluated the regional simulated runoff sensitivity of changing snowmelt conditions in the western U.S. using both Level III ecoregion boundaries (https://www.epa.gov/eco-research/ecoregions) and the MODIS derived SP zones (Hammond et al., 2017; Hammond et al., 2018). For each of the 10 major mountainous Level III ecoregions of the western U.S. evaluated by Barnhart et al. (2016), we compiled HRUs located within either the transitional and persistent snow zones (Figure 8). A statistical comparison between the means of each region compared to the mean of all regions using the non-parametric

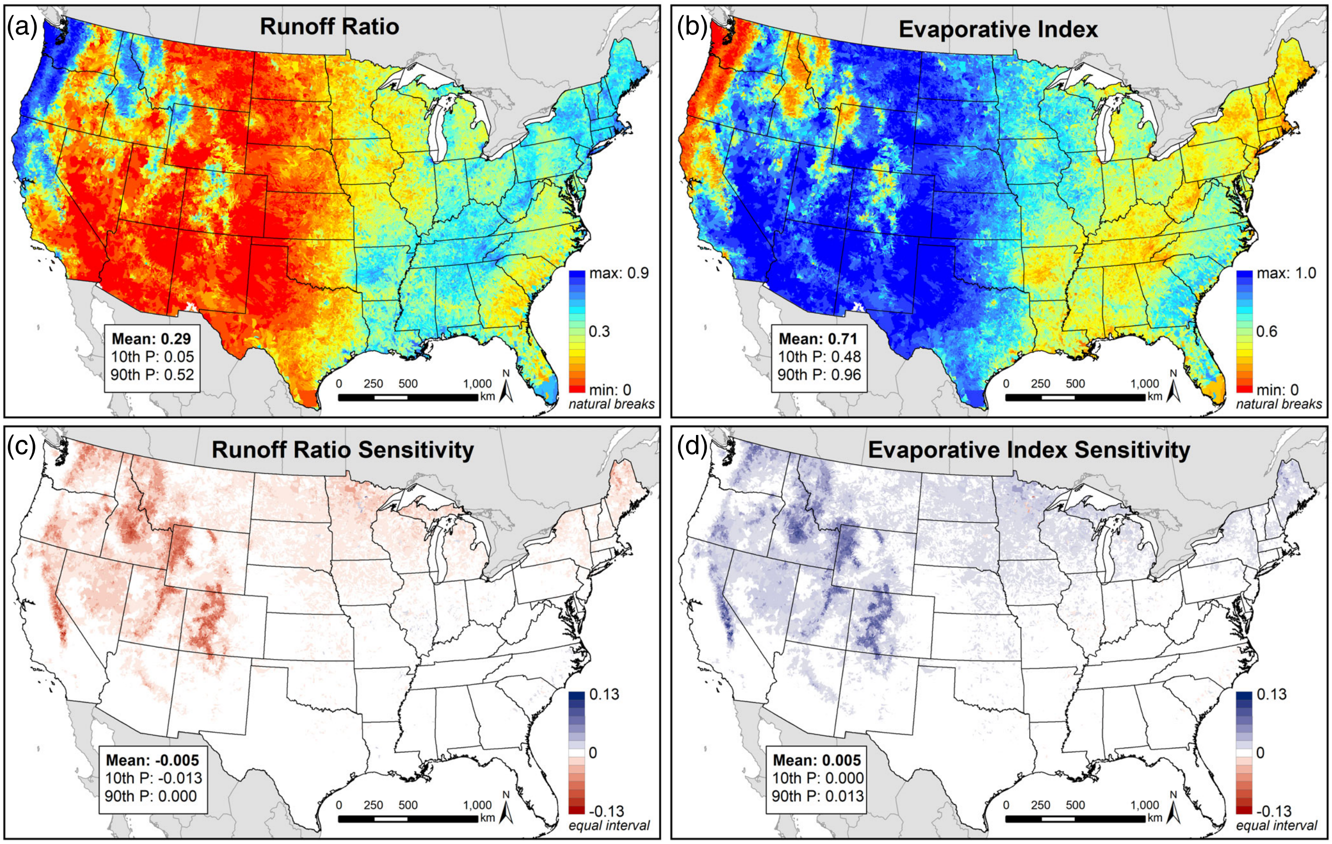

FIGURE 4 Spatial variability of the (a) runoff ratio and (b) evaporative index simulated for the coefficient of variation 0.5 model simulation and (c) the runoff ratio sensitivity and (d) evaporative index sensitivity across all model simulations. Mean value and 10th and 90th percentiles (P) across all hydrologic response units are shown in each panel

Wilcoxon rank sum test highlights that the Sierra Nevada, Idaho Batholith, Southern Rockies and Wasatch and Uintas ecoregions have significantly more runoff sensitivity and the Cascades, North Cascades, Northern Rockies, Canadian Rockies and Eastern Cascades ecoregions have significantly less runoff sensitivity to changing snowmelt compared to the mean of all regions (Figure 8). Figure 8 highlights this numbered order of regional sensitivity (from highest sensitivity to lowest sensitivity) to simulated decreases in snowmelt rate across the western U.S. mountains.

4 | DISCUSSION

4.1 | Sensitivity of simulated runoff to SDCs

Simulated runoff by the NHM-PRMS was shown to be widely sensitive to the SDC representation in snow-dominated regions of the CONUS. Across all model simulations, an increase in the derived SDC CV by 1.0 resulted in a runoff ratio decrease by as much as 12% and by 4% on average in the major mountainous ecoregions in the western U.S. (Figure 8). These decreases in runoff ratio were driven by slower simulated snowmelt rates averaged across the HRU, with decreases in snowmelt rate of 1 mm day^{-1} yielding decreases in runoff ratio of 1.8% on average (Figure 6a). Slower simulated snowmelt rates resulted in increases in simulated actual evapotranspiration promoted by an increase in the contact time of snow with the atmosphere as well as snowmelt water within the rooting zone. Our results indicate the importance of concentrated input to efficiently generating streamflow and are consistent with a recent modelling study by Barnhart et al. (2016) highlighting that faster snowmelt rates tend to promote greater streamflow generation by bringing the soil to field capacity, allowing for below-root zone percolation and streamflow generation. These results were further evidenced by recent 1-D soil input partitioning modelling (Hammond et al., 2019). Runoff ratio sensitivity to SDC representation presented in this study highlights the importance of carefully parameterising and evaluating SDCs for hydrologic modelling applications in snow-dominated regions.

4.2 | Scaling and model structure considerations

Our results showing that an increased sub-grid snow variability as represented by SDCs produces an effectively longer snowmelt duration and slower snowmelt rate at the HRU scale is supported by previous research (e.g., Luce et al., 1998). However, at the point scale, areas that hold snow longer through the snowmelt season, such as wind-induced snow drifts, are likely to experience faster snowmelt rates (Trujillo & Molotch, 2014), while also being associated with watersheds that have high sub-grid snow variability. Furthermore, many studies evaluating runoff in Reynolds Creek Experimental Watershed, Idaho, have highlighted that deep wind-induced snow drifts have high runoff efficiency and are a key source of runoff for the watershed (Chauvin et al., 2011; Flerchinger, Hanson, & Wight, 1996; Luce et al., 1998; Marshall et al., 2019; Stephenson & Freeze, 1974). Therefore, in the context of our results suggesting that runoff ratio decreases with increased snow variability at the watershed scale, an important scaling consideration is how hydrologic models distribute simulated snowmelt input across an HRU. Snowmelt input from the SCA simulated by the NHM-PRMS in this study was uniformly broadcasted across the entire HRU rather than being concentrated within the SCAs only. To test the effect of this model structure on simulated runoff response, we used PRMS to simulate a given volume of snow distributed unevenly across two HRU elements, and the same volume and snow distribution applied to a single HRU using an SDC (Figure 9). This "distributed" model using two HRUs versus "lumped" model using an SDC was tested based on a uniform snowpack with (a) 80% SCA and CV of 0.5 and (b) 50% SCA and CV of 1.1 (Figure 9). The climate and model parameters from a snow-dominated HRU within the Southern Rockies ecoregion was used for these test simulations over the same 14-year period as the original model experiment (Figure 8).

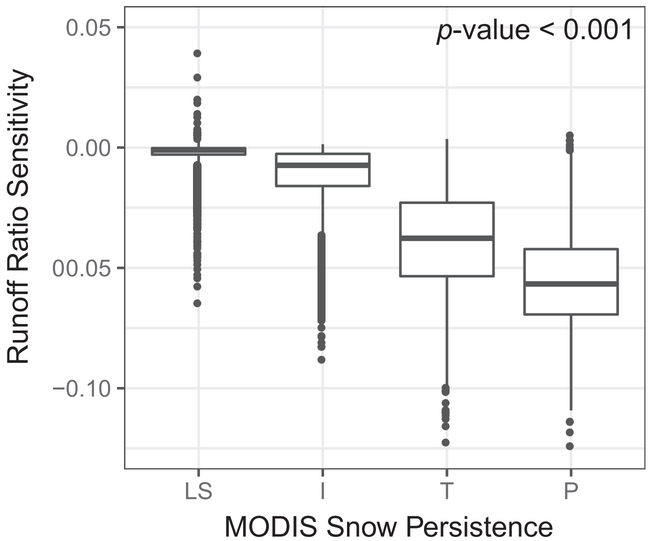

FIGURE 5 Boxplots of western U.S. runoff ratio sensitivity across all model simulations grouped by the snow persistence (SP) derived low snow (LS; SP ≤ 0.25), intermittent (I; 0.25 < SP ≤ 0.50), transitional (T; 0.50 < SP ≤ 0.75) and persistent (P; SP > 0.75) snow zones. Boxplots are hereinafter represented by the median (black horizontal line), 25th and 75th percentiles (box), and the largest (smallest) value within 1.5 times the interquartile range above (below) the 75th (25th) percentile (whiskers), and values that are >1.5 times and <3 times the interquartile range beyond either end of the box (outliers). The statistical significance (*p*-value) of difference between groups computed by the non-parametric Kruskal-Wallis test is shown

Figure 9 shows the simulated snowmelt rate, total snowmelt amount and runoff ratio during the snowmelt period for both the distributed (two HRUs) simulations and the lumped (SDC) simulations. Results highlight that simulated snowmelt rate and amount are comparable between the distributed versus lumped simulations when averaged across the entire HRU (Figure 9). However, when the snowmelt rate is averaged over the SCA only for the distributed simulations, the effective snowmelt rate was shown to increase substantially (Figure 9). Additionally, a clear effect in model structure is highlighted by the differences in simulated runoff ratio between the distributed

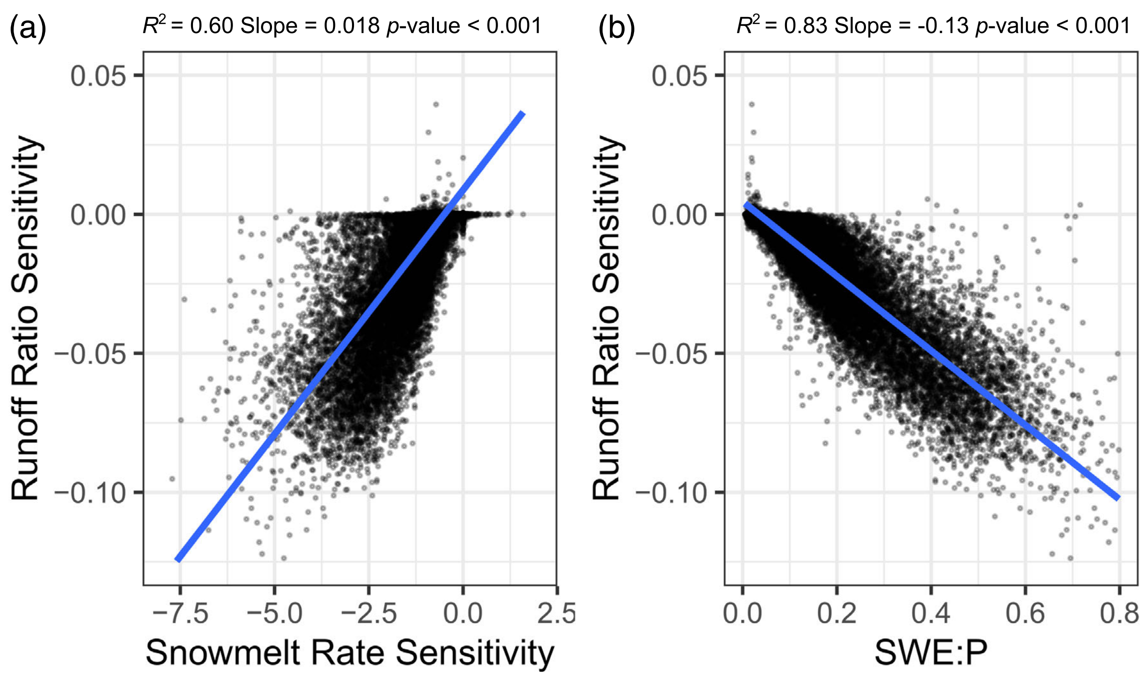

FIGURE 6 Scatterplots comparing both (a) the simulated snowmelt rate sensitivity across all model simulations and (b) the simulated peak snow water equivalent (SWE) over annual precipitation (P) ratio (SWE:P) to the western U.S. runoff ratio sensitivity across all model simulations

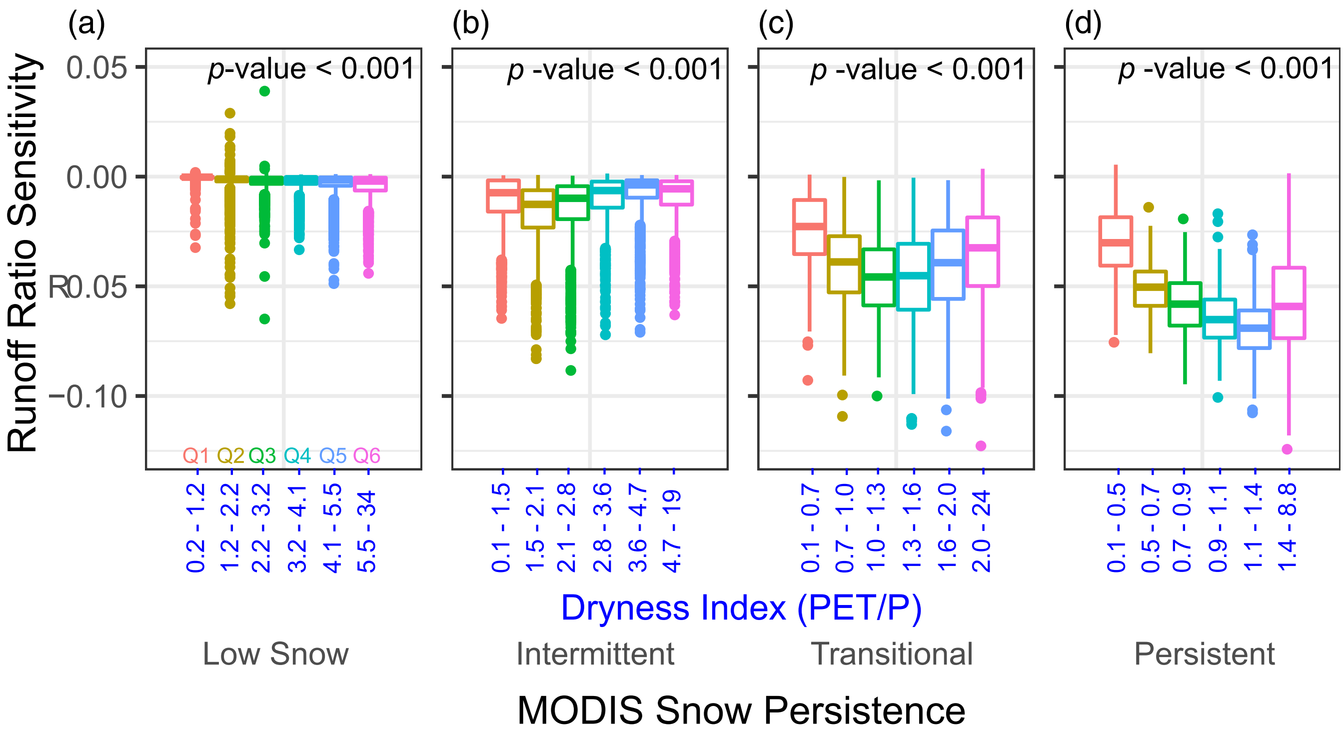

FIGURE 7 Boxplots of western U.S. runoff ratio sensitivity across all model simulations divided by six dryness index quantiles (coloured) computed for each of the (a) low snow, (b) intermittent, (c) transitional and (d) persistent snow persistence zones. The statistical significance (*p*-value) of difference between groups computed by the non-parametric Kruskal-Wallis test is shown on each panel

versus lumped simulations (Figure 9). The lumped simulations using an SDC to represent the snow variability resulted in a greater percentage of snowmelt being partitioned into evapotranspiration because the snowmelt water was uniformly distributed across the entire HRU rather than concentrated within the SCA (Figure 9). Furthermore, this effect was more pronounced for the CV of 1.1 (50% SCA) simulation compared to the CV of 0.5 (80% SCA) simulation (Figure 9). However, when taken together, the annual results from both the lumped and distributed simulations show a robust linear relation between snowmelt rate and runoff ratio (Figure 9; decreases in snowmelt rate of 1 mm day^{-1} result in decreases in runoff ratio of 3.6%) that is consistent with key findings from this study (Figure 6a; decreases in snowmelt rate of 1 mm day^{-1} result in decreases in runoff ratio of 1.8% on average in snow dominated regions of the CONUS).

These results indicate that accurately simulating the snowmelt input averaged across an entire HRU may not necessarily result in accurate simulation of snowmelt runoff generation. Recent studies showing that substantial snowmelt can be localized to specific areas of a watershed because of redistribution of meltwater and that greater runoff can be generated than if snowmelt were evenly distributed across landscape further highlight the importance of the sub-grid variability of snowmelt input (Eiriksson et al., 2013; Webb, Fassnacht, & Gooseff, 2018; Webb, Williams, & Erickson, 2018). Therefore, as suggested by Luce et al. (1998), an important challenge for future hydrologic modelling applications is relating the watershed-wide surface water input rate to runoff generation processes that are spatially dependent on snowmelt variability. Given the substantial effect of model structure (with respect to redistribution of snowmelt water across the HRU) to both the effective snowmelt rate and the associated hydrologic partitioning presented here (Figure 9), we suggest that additional research is needed in snow-dominated regions with dense data available for model evaluation to examine model improvements in representing sub-grid snowmelt input associated with the SDC and its relation with associated runoff generation processes. However, despite this substantial model scaling effect, we show that the relation and regional sensitivity between runoff generation and snowmelt rate presented in this study is robust to these model structure differences and provides important process insights; thus, we provide further discussion related to these results below.

4.3 | Runoff response to changing snowmelt dynamics

Observational evidence for slower snowmelt in warmer periods has led to modelling that shows meltwater produced at high snowmelt rates is greatly reduced in a warmer climate because of an earlier snowmelt season during a period of lower available energy (Musselman, Clark, Liu, Ikeda, & Rasmussen, 2017). The results presented in this study are important for providing insights to the CONUS-scale spatial variability of runoff sensitivity and response to changing snowmelt dynamics. However, the model experiment presented in this study does not represent any change in the timing of snowmelt initiation, which is also an observational result of changing

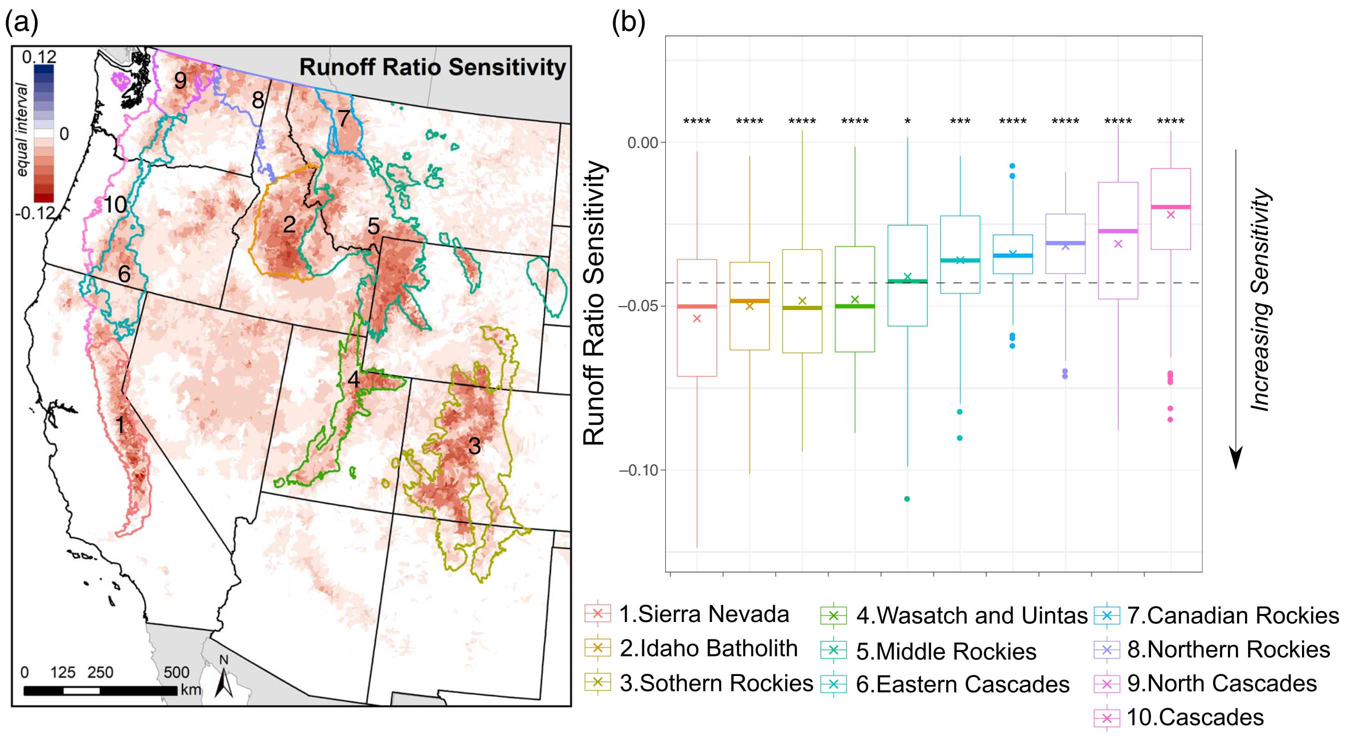

FIGURE 8 (a) The runoff ratio sensitivity across all model simulations shown for the western U.S. Level III ecoregions with the major mountainous ecoregions numbered for reference. (b) Boxplots of the sensitivity in simulated runoff ratio for the transitional (0.50 < SP ≤ 0.75) and persistent (SP > 0.75) snow persistence zones within each of the major mountainous ecoregions ordered from most sensitive to least sensitive (ordered left to right; numbered 1–10). The statistical significance of the difference of means between each region (denoted by the ×) with all regions (denoted by the horizontal dashed black line) computed by the non-parametric Wilcoxon rank sum test is shown above each boxplot (**** = *p*-value <0.0001; *** = *p*-value <.001; * = *p*-value <.1)

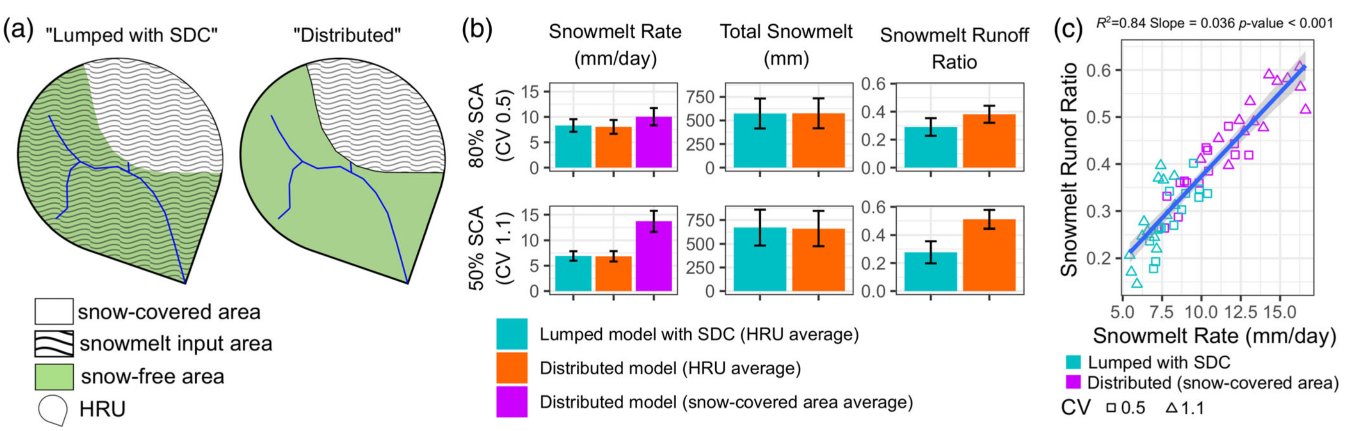

FIGURE 9 (a) Conceptual diagram showing the difference between the snowmelt input area across the hydrologic response unit for the "lumped with snow depletion curve" and "distributed" model configurations, (b) the resulting comparison of mean ± standard deviation snowmelt rate, total snowmelt, and runoff ratio during the snowmelt period for the distributed versus lumped model, and (c) scatterplot and linear relation comparing the simulated snowmelt rate to the simulated runoff ratio for both model configurations. Model comparisons are shown for a uniform snowpack distribution with (1) 80% snow-covered area (SCA) (coefficient of variation [CV] 0.5) and (2) 50% SCA (CV 1.1)

climate (Clow, 2010; Stewart et al., 2004) where the timing of snowmelt is also an important factor influencing the dynamics of evapotranspiration (Barnhart et al., 2016; Brooks et al., 2015; Harpold & Molotch, 2015; Molotch et al., 2009). In this study, higher evaporative indices consistently coincided with reduced runoff efficiencies (Figure 4), which is supported by previous work showing higher evapotranspiration in areas where snowmelt rate declines (Hammond et al., 2019). Other evidence indicates that decreased annual streamflow is also due to increased evapotranspiration in response to increased temperatures, earlier snowmelt and longer snow-free duration (Christensen & Lettenmaier, 2007; Foster, Bearup, Molotch, Brooks, & Maxwell, 2016; McCabe, Wolock, Pederson, Woodhouse, & McAfee, 2017; Woodhouse, Pederson, Morino, McAfee, & McCabe, 2016). The results from the model experiment in this study indicate that a decrease in snowmelt rate and increase in snowmelt season duration can result in increases in evapotranspiration that are approximately inversely proportional to decreases in runoff (Figure 4).

Distinct differences in the regional sensitivity of changes to snowmelt rate in the western U.S. mountains were observed in this study (Figure 8). Overall, the greatest simulated runoff ratio sensitivities were found in the persistent snow zone (Figure 5) where SWE:P is high (Figure 6). Although the simulated change in snowmelt rate was the driving mechanism resulting in a change in runoff ratio, a key result of this study is that the SWE:P was able to explain more of the variability in runoff ratio sensitivity than the change in snowmelt rate (Figure 6). This result highlights that the regional sensitivity of runoff to changing snowmelt rate is strongly linked to how much of the annual runoff originates as snowmelt (Li, Wrzesien, Durand, Adam, & Lettenmaier, 2017). For example, the transitional and persistent snow zones of the Sierra Nevada and Cascades ecoregions (Figure 8) both exhibited substantial decreases in simulated snowmelt rate during the model experiment (Figure 3b), but the SWE:P in the Sierra Nevada was much greater than the Cascades (Figure S4). Accordingly, the runoff sensitivity in the Sierra Nevada ecoregion was significantly greater than the Cascades ecoregion during the model experiment (Figure 8).

Furthermore, results from this study highlight how regional aridity is important to the sensitivity of runoff to changing snowmelt rate, with water-limited dryness index values between 1.0 and 1.6 in the transitional and persistent snow zones exhibiting the greatest runoff sensitivity (Figure 7). These results are important in the context of previous research that has shown that evaporative index varies along gradients of input seasonality, the fraction of precipitation falling as snow, input intensity and topography among other factors. With seasonal input, runoff may be more efficiently generated during the wet season with accompanying lower evapotranspiration efficiency, especially when subsurface storage is low (Carey et al., 2010). In areas with less seasonal precipitation and greater storage, evapotranspiration efficiency may increase. Similar to the results of our work, Berghuijs et al. (2014) show the highest sensitivity of hydrologic partitioning in the range of aridity index between 1 and 2. This may indicate that snowmelt dynamics play a greater role in hydrologic partitioning in this range than in very dry or very wet locations. Using the Budyko (1974) framework, Berghuijs et al. (2014) reveal greater positive streamflow anomalies with increasing fraction of precipitation falling as snow as well as corresponding decreases in evapotranspiration efficiency with increases in the fraction of precipitation falling as snow. Hammond et al. (2018) show that the effect of snow dominance on runoff efficiency is greater in cold/dry watersheds than warm/wet watersheds. Input intensity also strongly controls input partitioning to evapotranspiration, where efficiency declines with higher input rates (Barnhart et al., 2016; Wang & Tang, 2014) and greater input concentration in time (Hammond et al., 2019) in both arid and humid watersheds. The role of topography in controlling evapotranspiration efficiency depends primarily on slope, with steeper slopes limiting infiltrated input's time in the rooting zone (Voepel et al., 2011). A higher snowfall fraction (i.e., SWE:P) has been shown to be associated with a lower evaporative index (Berghuijs et al., 2014); therefore, results shown in Figure 7 highlight the combined influence of aridity and SP to runoff sensitivity.

4.4 | Implications for future modelling applications

The model experiment presented in this study was designed specifically to evaluate the sensitivity of simulated runoff to the representation of SDCs within the model rather than determining the most accurate SDC representation across the CONUS scale. However, results presented here provide important insights into future development of model representations of snow variability within the NHM-PRMS and other models using methods to represent subgrid snow variability (e.g., Liston, 2004; Newman, Clark, Winstral, Marks, & Seyfried, 2014; and references within). Overall, Figure 4 highlights the HRUs with simulated runoff that are particularly sensitive to SDC representation within the model, which are generally situated in the persistent and transitional snow zones (Figure 5). The development and calibration of SDCs for these HRUs may warrant special attention and potentially prioritization. Additionally, this study highlights the importance of representing the sub-grid variability of snowmelt input into a watershed (Figure 9). Simulating the sub-grid SWE variability and resulting snowmelt input averaged across an HRU may not necessarily result in accurate simulation of snowmelt runoff generation; therefore, model improvements to represent the sub-grid snowmelt input associated with the SDC and its relation with associated runoff generation processes are needed.

The spatially variable dimensionless SDCs based on the lognormal pdf (Figure 1) presented in this study could be implemented in future applications across the CONUS using the SWE CV classification system developed by Liston (2004). Using these SDCs as a starting point, targeted calibration strategies (based on the sensitivity results presented in this study) to remotely sensed SCA observations could then be completed (Hay, 2019). Calibration could then be evaluated using both SWE and SCA observations from both remote sensing platforms and ground-based observations (e.g., Arsenault & Houser, 2018; Fassnacht et al., 2016) as well as results from fine-scale snow modelling applications (e.g., Broxton, van Leeuwen, & Biederman, 2019; Hedrick et al., 2018; Sexstone et al., 2018). Additionally, known relations between SDCs and topography and climatic variables (Driscoll et al., 2017) could be utilized for evaluation. Potential scaling challenges and uncertainties of these methods remain in that Pomeroy et al. (2004) and DeBeer and Pomeroy (2010) show that a lognormal distribution can fail to accurately represent snow distributions when aggregated over varying landscape classes. This indicates there may be a need to further disaggregate HRUs in the NHM-PRMS to more accurately represent sub-grid snow processes; further investigation is needed to evaluate the trade-offs between computational efficiency, data availability and improved process representation within this national-scale modelling system (e.g., Archfield et al., 2015).

5 | CONCLUSIONS

In this study, we applied a daily time-step, deterministic, distributed-parameter, process-based hydrologic model across the extent of the CONUS to investigate the sensitivity of simulated hydrologic component fluxes to the representation of SDCs within the NHM-PRMS. Increases in the sub-grid variability of SWE as represented by SDCs in the model resulted in a consistently slower snowmelt rate and longer snowmelt duration when averaged across the HRU scale. As a result of slower snowmelt rates, simulated runoff ratios were shown to decrease in snow-dominated regions by 1.8% on average with associated decreases in snowmelt rate by 1 mm day^{-1}. Increases in evaporative index were approximately inversely proportional to decreases in runoff ratio, indicating decreases in runoff at small scales may primarily be attributed to increasing evapotranspiration because of an increase in the contact time of snow with the atmosphere as well as snowmelt water with the rooting zone. However, across the large HRUs in this model application, hydrologic partitioning was shown to be sensitive to fast snowmelt from small SCAs being effectively slowed by being redistributed across the entire HRU area. Regions of the CONUS with high SP and SWE:P exhibited the greatest runoff sensitivity to changes in snowmelt rate. Within these snow-dominated regions, runoff sensitivity was greatest in semi-arid areas with dryness index values between approximately 1 and 2 (water limitation). These results provide important insights into the CONUS-scale spatial variability of runoff sensitivity to changes in snowmelt dynamics and highlight the importance of carefully parameterizing SDCs within hydrologic modelling. Improving model representation of snowmelt input variability and its relation to runoff generation processes is shown to be an important consideration for future modelling applications. Furthermore, this study provides guidance for future development of the NHM-PRMS to improve modelling and predictive capacity of water availability across the CONUS.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available in USGS ScienceBase at https://doi.org/10.5066/P9OEIRJF.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of this article.

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